LAB 11

ML-Signal Stage: a Walk-Forward Classifier from Probability to Position

Item	Detail
Topic	Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals
Pipeline stage	ML signal
Duration	25 min
Simulator page	http://localhost:8000/#/research/ml
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

The desk's data scientist claims a machine-learning model reading technical features can predict whether SPY closes up tomorrow. The pipeline will only accept an ML signal that is validated walk-forward, turned into positions with an explicit probability threshold and traded next bar with the same costs as every other signal - and that still beats buy and hold.

Objective

Train logistic regression, random forest and XGBoost walk-forward and trade P(up) > threshold.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 11. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 11: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

Step 1. Train the classifiers

Research > ML models on SPY, threshold 0.55.

Open: http://localhost:8000/#/research/ml (Research > ML SPY)

Check: AUC close to 0.5 - daily direction is close to a coin toss; read the verdict.

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Step 2. Backtest an ML strategy

Backtest 'ML: XGBoost (walk-forward)' on SPY.

Open: http://localhost:8000/#/backtest?strategy=ml_xgboost&symbols=SPY (Backtest SPY XGBoost)

Check: Judge the model by the net-of-cost strategy, not by accuracy.

Step 3. Check for leakage

Research > Diagnostics on 'ML: logistic regression'.

Open: http://localhost:8000/#/research/diagnostics?strategy=ml_logistic

Check: Look-ahead check PASS: refits use only past bars.

Expected in the app: Accuracy / AUC table, feature importance and a net-of-cost backtest of the ML signal.

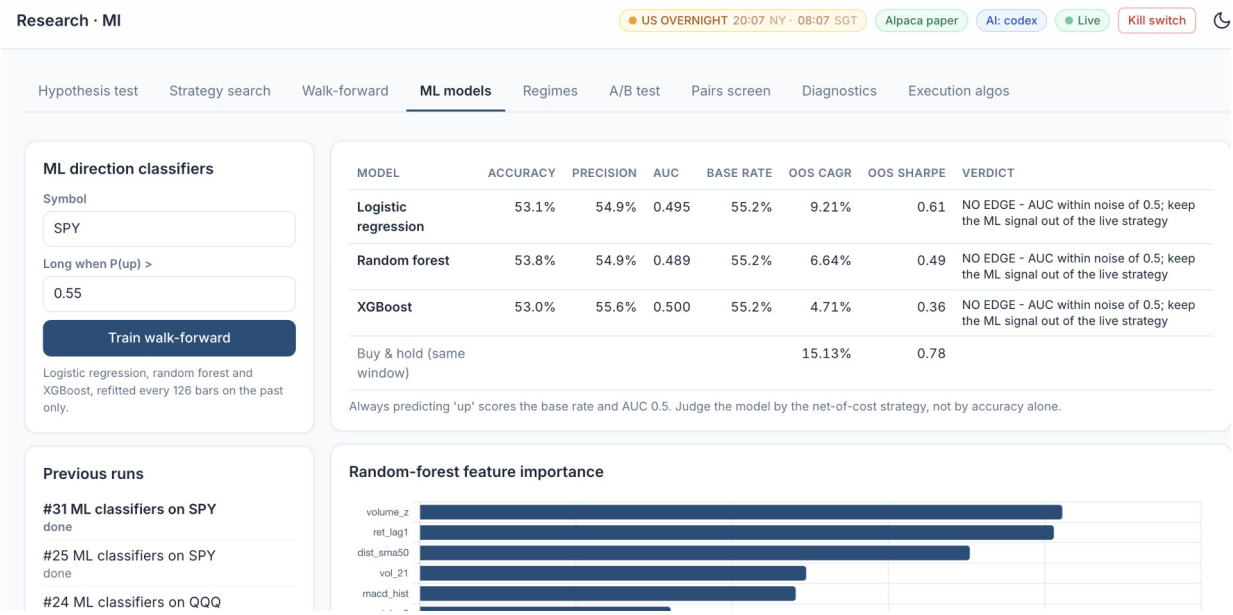


Figure: Lab 11 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

A walk-forward machine-learning signal - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: predict whether SPY closes higher tomorrow with walk-forward logistic regression and random forest classifiers (expanding window, first 504 days to train, retrain every 126 days), using features computed with data up to today only: RSI(14), 5- and 21-day returns, 21-day volatility and the distance from SMA(50). Report accuracy and AUC per model and the feature importance, and backtest 'long when P(up) > 0.55' (trade at the next day's open with a 0.05% cost per trade) against buy-and-hold.

Open in Claude | Open in ChatGPT | Open Google Colab

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Debrief

Supervised models can be trained and validated correctly and still find no edge - an honest negative result is valuable evidence. Walk-forward validation stops the model learning from the future, a leakage test proves the check works, and the cost-aware backtest shows what an AUC of 0.5 is worth in money.

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